

Mohammed Nagi

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RESEARCH PROFILE

Google DeepMind Scholar and MSc Artificial Intelligence, trained originally as an electrical and electronics engineer. My research imports tools and intuitions from signal processing and physical systems modelling into mechanistic interpretability. Across my work, I prioritise claims that can be falsified by intervention over claims supported only by correlation, and I am motivated by the broader question of what studying intelligence in physically grounded, non-linguistic systems can teach us about interpretability methods built primarily on language models.

EDUCATION

African Institute for Mathematical Sciences (AIMS)

Master of Artificial Intelligence (Google DeepMind Scholar)

Sep 2025 – July 2026

South Africa

– **Grade:** Distinction

– **Relevant Coursework:** Natural Language Processing & LLMs, Applied Machine Learning at Scale, Explainable AI (XAI), Reinforcement Learning, Responsible AI, CUDA programming for NVIDIA GPUs, Machine Learning Operations (MLOps).

University of Khartoum

Bachelor of Electrical and Electronics Engineering (Honours)

2017 – Aug 2025

Sudan

– **Grade:** First Class Honours

– **Context:** Maintained academic continuity and research output despite severe regional conflict and infrastructure collapse.

– **Relevant Coursework:** Calculus, Linear Algebra, Systems Modelling and Simulation, Signal Processing, Real-Time Embedded Systems.

RESEARCH & TECHNICAL PROJECTS

MSc Thesis: Sparse Autoencoders for Interpretability of InstaNovo | *Python, PyTorch* July 2026

- Trained Sparse Autoencoder dictionaries (12,288 features) on the residual-stream activations of InstaNovo, a transformer that encodes physical mass-spectrometry signals rather than language and reconstructed activations with 92–98.4% fidelity while preserving over 99.5% of downstream sequencing loss on 639,286 spectra.
- Built a first-principles fragment-ion annotation pipeline deriving 50 chemical concepts per spectral peak directly from fragmentation physics, then tested feature-concept association with a dominance-weighted F1 score, inspired by prior SAE work on protein language models (*Simon & Zou, 2025*).
- Ran firing-rate-matched causal ablations to separate correlation from function: the sharpest correlational feature was causally inert on ablation, while a modestly correlated y-ion feature group was catastrophic to remove, demonstrating a structural bias in correlation-based feature ranking.

Fairness-Constrained Bandit Routing for Human-AI Triage | *PyTorch, SHAP, Fairlearn* 2026

- Architected PSYCHROUTE, a fairness-constrained human-AI collaboration framework for psychiatric triage (Bipolar Disorder vs Major Depressive Disorder), wrapping a fine-tuned MentalBERT classifier in a LinUCB contextual bandit that routes each encounter across four governance-anchored actions mapped to the Fabric AI governance taxonomy (*Jorgensen et al., 2025*).
- Achieved a 35% reduction in diagnostic error rate (0.017 vs AI-only 0.026 and human-only 0.070) on 3,626 held-out encounters, with formal verification of human-AI complementarity in both directions (*Mozannar & Sontag, 2020*).

Explainable AI (XAI) & Mechanistic Interpretability | *PyTorch, SHAP, LIME, TCAV* 2026

- Implemented axiomatic Shapley decomposition, LIME, and counterfactual explanations from scratch, then contrasted their correlational attributions against Integrated Gradients and Testing with Concept Activation Vectors (TCAV) on Vision Transformers and sequence models, building direct intuition for where attribution-based explanation methods and causal/mechanistic methods agree and diverge.

AI-Driven Discovery in Spectral Graph Theory | *Python, PyTorch, RL, GNNs* 2026

- Extended the Cross-Entropy Method (*Wagner, 2021*) with Graph Neural Network policy approximators to discover counterexamples to open conjectures in spectral graph theory, automating search through discrete combinatorial spaces (graphs up to $N = 35$) to surface structural violations of established mathematical bounds.

TECHNICAL SKILLS

Languages: Python, CUDA, SQL, MATLAB, LaTeX

ML & Interpretability: PyTorch, Scikit-learn, TransformerLens, Sparse Autoencoders, SHAP, LIME, Integrated Gradients

Causal & Statistical Methods: Causal Interventions, Feature Ablation, Bayesian Inference, Monte Carlo Methods

Safety, Alignment & Fairness: MC-Dropout Uncertainty, Temperature Calibration, Fairlearn Auditing, Contextual Bandits (LinUCB)

Tools & Platforms: Git, Docker, Google Cloud Platform (GCP)

Research Interests: Mechanistic Interpretability of Non-Linguistic and Physically-Grounded Models, AI Safety, Human-AI Collaboration

Natural Languages: Arabic (Native), English (Fluent), French (Intermediate)